

Rank Before You Merge: Native-Unit Selection for Visual Token Compression

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Abstract

Visual token compression depends on both the retained budget and the representation used to select it. We study this dependence in vision-language models with native spatial mergers. Rank-Before-Merge (RBM) scores complete native units using pre-merger feature norms and keeps their identities aligned with the model’s visual streams, without training or changing merger weights. At 25% retention, RBM exceeds post-merger norm selection on all nine text-dense model–benchmark pairs across Qwen3-VL, Qwen2.5-VL, and InternVL3. These operational comparisons include model-specific feature taps and allocation rules. To reduce the feature-depth confound, a separate full-split Qwen3-VL final-input control scores the main merger’s own input: it improves TextVQA by 27.68 percentage points, DocVQA by 4.59 points, and OCRBench by 235/1000, but reduces GQA by 5.64 points. The post score still aggregates main and auxiliary outputs, so this control is not a pure intervention on merger order. Rank disagreement and edge-energy associations characterize the selected evidence; a ranking-swap check verifies path equivalence for a fixed kept set. Comparisons with FastV show complementary task preferences rather than uniform superiority. Code: github.com/Code-Yan-ZX/qrbm-vlm.

Introduction

High-resolution images produce long visual sequences in modern vision-language models (VLMs). A common architectural response is a native spatial merger: neighboring patch features are converted into fewer language-model inputs. Additional compression must then decide which of these spatial units deserve a smaller token budget. This decision depends on the scoring representation. A large norm before a learned projection need not remain large afterward, and a downstream selector cannot recover evidence from a unit it discards.

We investigate this choice using a deliberately simple, query-blind norm score. Rank-Before-Merge (RBM) selects complete native units from pre-merger features while retaining native merger weights and downstream interface dimensions. It requires neither training nor a second learned compression module. Its purpose is to expose a useful selection signal while respecting the model’s existing spatial topology.

Two questions must be separated. First, does an operational RBM implementation preserve accuracy at the same final token budget as post-merger norm selection? Second, does a later pre-merger tap retain the same direction of effect after removing

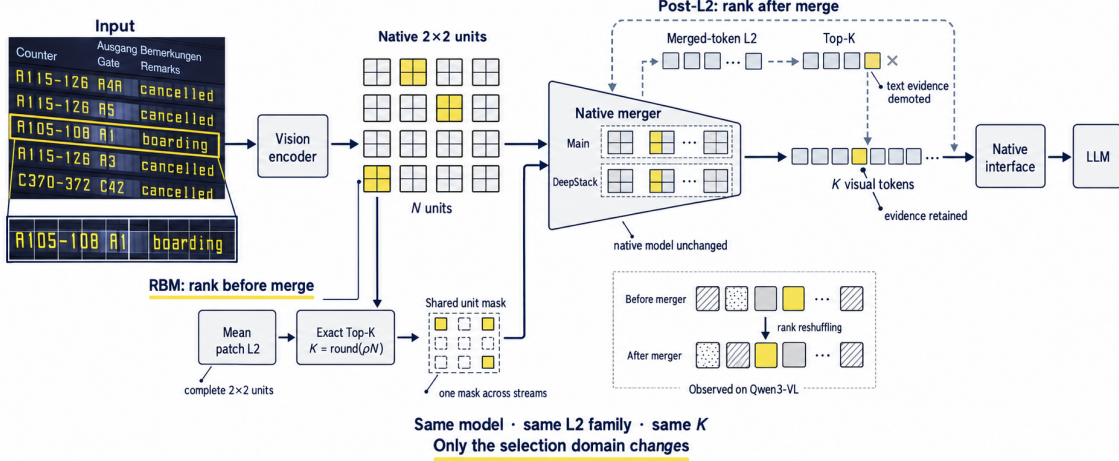


Figure 1: Native-unit selection schematic. Selected identities stay aligned with the surviving visual streams and interface. The illustrated pre-merger location is conceptual: operational Qwen3-VL RBM scores the layer-8 deepstack input, whereas the final-input control scores the main merger’s final-block input. The diagram’s ρ denotes retention, written κ in the text.

the early-feature advantage? Our three-model comparison addresses the first question. A separate Qwen3-VL experiment scores the main merger’s final-block input while holding sample identities and token counts fixed. It reduces, but does not eliminate, implementation differences because the post score concatenates main and auxiliary-stream outputs. This distinction matters because operational Qwen3-VL RBM scores an earlier tap and prunes multiple visual streams.

At 25% retention, operational RBM improves all nine text-dense comparisons against Post-L2 across three model families. The final-input control also favors its pre-merger score on TextVQA, DocVQA, and OCRBench, but the DocVQA gain is substantially smaller. GQA reverses the direction. Thus neither an architecture-independent preference for earlier features nor a universal advantage over stronger selectors follows from the results.

Our contributions are a native-unit compression rule compatible with deployed merger interfaces; a full-split evaluation separating operational performance from a stricter final-input control; and diagnostics and attention-based comparisons that identify the scope of the observed advantage. We interpret visual-token retention as a task compression budget, without equating it to a bit rate or to total inference cost.

Related Work

Visual-token reduction can operate inside the language model. FastV uses early-layer attention to select visual tokens [1]; PyramidDrop, FasterVLM, and SparseVLM explore layerwise or attention-based reduction [2, 3, 4]. Their question-conditioned signals differ from RBM’s image-only norm score. A comparison at equal final token count is informative about accuracy, but does not imply equal computation because the candidates can remain active through different numbers of layers.

A second line reduces or aggregates visual features before language processing. ToMe merges similar tokens [5]; LLaVA-PruMerge and VisionZip combine saliency and redundancy reduction [6, 7]; TokenPacker learns a compact visual projector [8]. FitPrune estimates pruning allocation, and AdaptMerge uses language guidance [9, 10]. These methods motivate studying representation and selection jointly, but our experiments do not rank this entire design space.

We focus on architectures with native spatial compression: Qwen2.5-VL, Qwen3-VL, and InternVL3 [11, 12, 13]. RBM retains their native operators and selects complete input units. Post-L2 is a diagnostic baseline from the same feature-magnitude family; FastV supplies a stronger attention-based comparison. The resulting evidence concerns these model implementations and workloads, rather than a claim that downstream selectors generally assume rank preservation.

Native-Unit Compression

Selection representation and token budget

Let an image contain N native spatial units, each grouping four patch vectors. At retention κ , retain

$$k = \max\{1, \text{round}(\kappa N)\}. \quad (1)$$

For features $h_{u,j}^{(t)}$ at tap t , RBM uses

$$a_u^{(t)} = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}^{(t)}\|_2, \quad K_{\text{RBM}} = \text{TopK}_k\{a_u^{(t)}\}. \quad (2)$$

Post-L2 instead scores a returned visual vector z_u :

$$b_u = \|z_u\|_2, \quad K_{\text{post}} = \text{TopK}_k\{b_u\}. \quad (3)$$

Both rules use magnitude, but they evaluate different representations; they are not identical scoring functions applied to identical vectors. In Qwen3-VL, z_u concatenates the main and three deepstack outputs, whereas the other tested paths use their model-specific post-projector vectors.

In the idealized single-stream case with $h^{(t)}$ at the merger input, $z_u = M(h_{u,1:4}^{(t)})$. For a per-unit merger and a fixed kept set, selection can be performed before or after M without changing the retained outputs. In contrast, the ranking decisions coincide only when their top- k sets agree. A strictly increasing map $b_u = g(a_u)$ is sufficient for agreement at all budgets, but is not a property we assume of learned mergers. We quantify set agreement with Jaccard overlap, $|K_{\text{pre}} \cap K_{\text{post}}| / |K_{\text{pre}} \cup K_{\text{post}}|$. Low overlap alone predicts neither which set is more useful nor the size of an accuracy difference. Qwen3-VL’s concatenated post vector departs from this single-stream idealization, so its final-input experiment is a stricter control rather than exact causal isolation.

Model-specific implementation

RBM retains complete units and keeps selected features aligned with the positions and auxiliary streams expected by each adapter. Model weights, merger weights,

interface dimensions, and decoding are unchanged. The implementations follow their backend’s compressed-position convention; this does not imply that every backend gathers the selected units’ original spatial coordinates. L2 scoring costs $O(Nd)$ for feature dimension d ; top- k can be implemented in $O(N \log k)$ with a heap. The method adds no learned parameters or language-model attention pass.

For Qwen3-VL, operational RBM scores the first deepstack input at ViT layer 8 and applies one kept set to the main and three deepstack mergers. Post-L2 scores the concatenated returned vector. Consequently, their operational comparison changes feature depth, score aggregation, and where auxiliary streams are pruned. Qwen2.5-VL scores the main merger input, but its available pre path keeps units in window order while the post path follows the native output restoration. InternVL3 scores the concatenated four-patch vector after parameter-free pixel shuffle and before its learned projector; RBM allocates a quota per tile, whereas Post-L2 pools each image’s tiles before top- k . These results are therefore operational accuracy comparisons, not isolated cross-model merger effects.

The separate Qwen3-VL *Pre-final* control scores final ViT-block features at the main merger input. Pre-final and Post-L2 both execute the deepstack mergers in full and select corresponding rows afterward. Their configuration, sample identities, and per-image token counts match. This removes the early-tap difference, but the pre score still uses four input patch norms while Post-L2 uses a concatenated main-plus-deepstack vector. We therefore call it a final-input control, not a pure stage intervention. It is also not an efficiency measurement of physically skipping merger work, and RBM–Pre-final differences are not pure feature-depth effects.

Evaluation Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B with bf16 weights and greedy decoding. The main matrix and final-input control use vLLM 0.19 V1 [14]. FastV and the hybrid probe use HuggingFace eager attention. We compare methods within each backend; small checks matching native outputs on 8/8 examples and HF/vLLM outputs on 16/16 examples do not establish full-split backend equivalence.

The full evaluation sets contain 5,000 TextVQA examples, 5,349 DocVQA examples, 1,000 OCRBench examples, and 12,578 GQA examples. Metrics are VQA accuracy, ANLS, OCRBench score out of 1,000, and normalized exact match, respectively [15, 16, 17, 18]. Scoring follows the corresponding lmms-eval rules [19]. The three former benchmarks are designated text-dense; GQA provides an object-centered question-answering contrast.

All reported compression comparisons retain 25% of native units and match per-image final visual-token counts within each experiment. Qwen3-VL final-input runs use native image resolution, at most 32 generated tokens, and eager vLLM execution. TextVQA, OCRBench, and GQA use a maximum sequence length of 8,192 and up to eight concurrent sequences; DocVQA uses 32,768 and up to four. Both control arms complete every evaluation sample. The reused uncompressed Qwen3-VL anchors score two TextVQA and 18 OCRBench context overruns as zero. Qwen2.5-VL’s operational OCRBench pair uses a shared four-million-pixel cap. InternVL3 DocVQA

Table 1: Full-split operational results at 25% retention. None is uncompressed. Qwen3-VL RBM uses the early tap; this table does not isolate scoring stage. Deltas use unrounded scores and are percentage points except for OCRBench (/1000).

Model	Benchmark	None	RBM	Post-L2	Δ
Qwen3-VL	TextVQA	.844	.605	.222	+38.4
	DocVQA	.956	.481	.238	+24.3
	OCRBench	760	547	184	+363
	GQA	.616	.449	.477	-2.8
Qwen2.5-VL	TextVQA	.862	.702	.442	+26.1
	DocVQA	.949	.636	.526	+11.0
	OCRBench	817	480	182	+298
	GQA	.604	.559	.585	-2.6
InternVL3	TextVQA	.8338	.7890	.4148	+37.4
	DocVQA	.9221	.7284	.3820	+34.6
	OCRBench	852	753	321	+432
	GQA	.6293	.5993	.6031	-0.4

also uses a four-million-pixel cap. Absolute scores across model families are therefore not controlled model rankings.

For the final-input control, 95% confidence intervals use 20,000 paired-bootstrap resamples of per-example official scores. We report these intervals separately from the exploratory hybrid probe, for which we make no significance claim. Full-split scores, fixed-size probes, and skipped-example handling are distinguished below.

Results

Operational accuracy across model families

Table 1 compares the deployed RBM design with Post-L2. RBM improves all nine text-dense model-benchmark pairs: 11.0–38.4 percentage points on TextVQA and DocVQA, and 298–432 points on OCRBench. It nevertheless remains lossy relative to the uncompressed models. For example, Qwen3-VL DocVQA falls from .956 to .481, despite its advantage over Post-L2. A favorable relative comparison is not equivalent to preserving the original model’s accuracy.

GQA favors Post-L2 in all three point estimates. The small InternVL3 difference has an interval crossing zero in the paired analysis. The Qwen results give clearer evidence of a task boundary: pre-merger magnitude is useful for the tested text-dense workloads, but is not a general-purpose saliency criterion.

Final-input control

Table 2 and Fig. 2 separate the Qwen3-VL final-input contrast from the operational early-tap design. Pre-final improves TextVQA by 27.68 points and OCRBench by 235/1000. DocVQA improves by 4.59 points, compared with the 24.3-point operational contrast. These are distinct effects; attributing the latter entirely to the main merger boundary would overstate the evidence. Conversely, GQA decreases by 5.64

Table 2: Qwen3-VL final-input control, complete full splits. Pre-final scores four input patches; Post-L2 scores the concatenated main-plus-deepstack output. Deltas and 95% paired-bootstrap intervals are percentage points, except OCRBench points.

Benchmark	Pre-final	Post-L2	Δ [95% CI]
TextVQA	.4985	.2217	+27.68 [26.21, 29.19]
DocVQA	.2836	.2377	+4.59 [3.43, 5.78]
OCRBench	419	184	+235 [199, 271]
GQA	.4207	.4771	-5.64 [-6.42, -4.86]

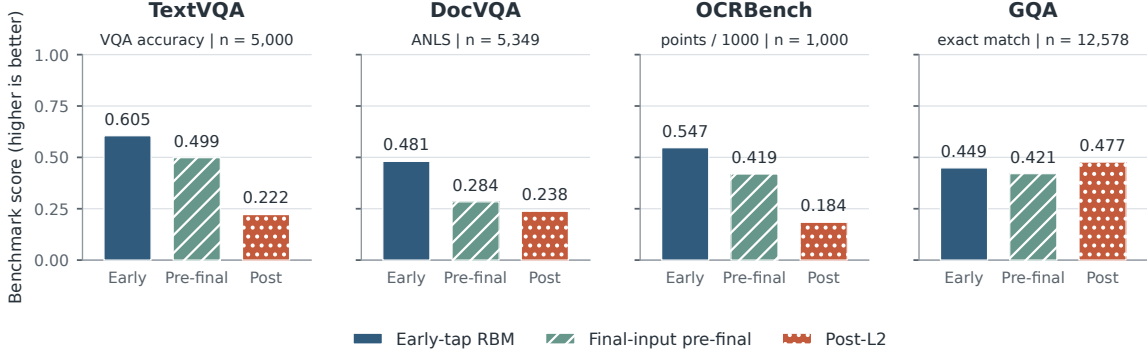


Figure 2: Full-split Qwen3-VL accuracy at 25% retention. Early-tap RBM is the operational design; Pre-final and Post-L2 form the stricter final-input comparison. OCRBench is normalized by 1,000. The RBM-Pre-final difference also includes auxiliary-stream handling and is not a feature-depth-only intervention.

points in the final-input comparison. The full-split control therefore shows that a later pre-merger representation retains a workload-dependent effect, while its residual score-aggregation difference limits a merger-only attribution.

Comparison with attention-based selection

A same-harness full-OCRBench comparison evaluates RBM and FastV-k3 at a shared four-million-pixel cap and a 25% final budget (Table 3). Both methods use native positions, greedy decoding, and at most 32 generated tokens. The Qwen3-VL and Qwen2.5-VL runs encounter 79 and 88 common context or memory failures, respectively. These exact shared IDs receive zero in the scores out of 1,000; the completed paired sets contain 921 and 912 examples. RBM leads by 146 and 87 points. These HF scores are a separate experiment and must not be substituted into the vLLM matrix.

We also examine a hybrid, RankBridge, that protects $q = \text{round}(\alpha k)$ RBM-ranked positions and fills the remaining $k - q$ positions from FastV’s ranking, excluding protected positions. Its prior weight $\alpha = 0.2$ was selected from $\{0.1, 0.2, 0.3\}$ in a development gate. All candidates remain available until the FastV selection layer, $K = 3$. Thus the hybrid’s equal final token count does not give it RBM’s earlier sequence reduction.

Table 4 reports the resulting exploratory Qwen3-VL probe. TextVQA, DocVQA,

Table 3: Same-harness HF eager OCRBench comparison, 1,000 attempted examples per model. Common failures are scored zero; both methods retain the same final token count.

Model	RBM	FastV-k3	Δ (/1000)	Completed
Qwen3-VL	559	413	+146	921
Qwen2.5-VL	382	295	+87	912

Table 4: Exploratory Qwen3-VL hybrid probe at 25% final retention. n is the common effective sample size; OCRBench is accuracy on the 181 completed samples. Bold marks the highest observed score.

Benchmark	n	RBM	FastV-k3	RankBridge
TextVQA	200	.5967	.7633	.7950
DocVQA	200	.4239	.5863	.6023
OCRBench	181	.5801	.4586	.4641
GQA	200	.4150	.5050	.5100

and GQA use 200 examples each. OCRBench has 181 jointly evaluable examples after 19 common loading skips; its row reports accuracy over those 181 examples. RankBridge improves the observed FastV scores by 0.5–3.2 percentage points, but trails RBM on OCRBench by 11.6 points. It fails the development objective of remaining within one point of RBM there. The available results suggest complementary signals, not a validated universal hybrid. We make no claim of a statistically significant improvement or of an independently held-out confirmatory test.

Diagnosics and Interpretation

The operational Qwen3-VL pre/post ranks differ substantially. On the diagnostic DocVQA, TextVQA, and GQA samples, Spearman correlations are .137, .332, and .360, and top-25% Jaccard overlaps are .180, .243, and .278. These observations show that representation choice changes the selected set. Because the operational ranks use different feature depths, these diagnostics alone do not isolate the main merger’s contribution.

Rank shifts also associate with local edge energy. Correlations between post-minus-pre rank and Sobel energy are +.439, +.155, and +.036 for the same task order. On DocVQA, RBM-only units have mean edge energy .641 versus .124 for Post-L2-only units; 92% versus 35% exceed their image’s median. Sobel energy responds to borders, texture, and object contours as well as text. The measurements therefore establish an edge association, not text-specific demotion or a causal explanation of answer accuracy. Figure 3 illustrates different spatial allocations without serving as aggregate evidence.

A ranking-swap check fixes the post-merger computation but substitutes RBM’s kept identities. On the verified Qwen3-VL path, its kept-set overlap with RBM is 1.000, retained representations are byte-identical, and predictions agree. This verifies path equivalence when the set is fixed and excludes altered computation on surviving

Kept visual units (blue) per selector — two OCR cases

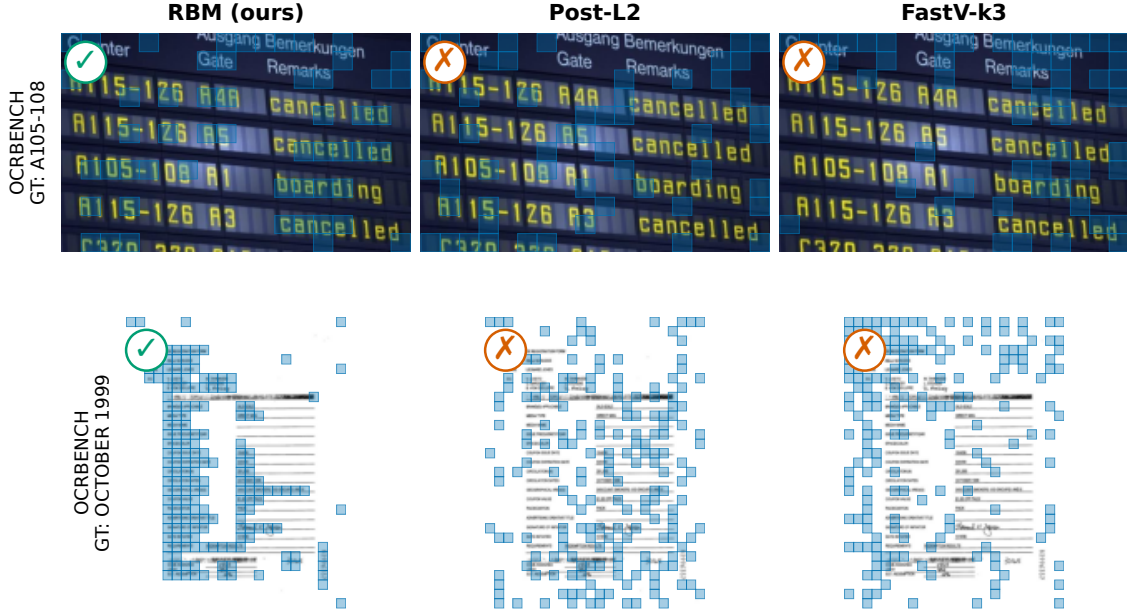


Figure 3: Illustrative OCRBench selections at 25% retention. RBM, Post-L2, and FastV-k3 preserve different spatial regions. These examples visualize selection behavior; quantitative comparisons use the separately specified evaluation sets.

units as the explanation for that check. It does not establish that norm or edge energy identifies causally relevant text. The analogous Qwen2.5-VL path retains an ordering confound; InternVL3 supplies accuracy-level replication, without an equivalent byte-exact attribution claim.

Cost and Limitations

The retained-unit fraction k/N measures the final visual sequence budget. It is neither an encoded bit rate nor a sufficient measure of inference cost. RBM, Post-L2, and FastV differ in the stage at which sequences are reduced. In particular, FastV and RankBridge process all candidate visual tokens through the initial language-model layers. Native image resolution, output length, batching, and backend also affect wall-clock performance. The method does not eliminate vision-transformer work before the scoring tap, and the Pre-final control deliberately executes the full merger streams.

On one A40, a Qwen3-VL TextVQA batch of 200 examples reaches 6.91 requests/s with RBM at 25% retention, versus 4.11 without compression and 6.89 with Post-L2. The vLLM run uses native image resolution, at most eight concurrent sequences, greedy decoding capped at 32 output tokens, and one untimed warmup; timing surrounds batched generation and excludes engine construction and initial image loading. A separate five-run, single-example proxy averages .23 seconds for RBM and .36 seconds without compression. The throughput result supports an end-to-end gain for this workload, while the nearly equal RBM/Post-L2 rates do not show a selector-specific

speed advantage. The one-example proxy is not a serving-latency distribution.

Our strongest controlled conclusion is the full-split Qwen3-VL final-input comparison, with the stated score-aggregation limitation. The three-model operational matrix extends the empirical pattern across native merger designs, but does not establish a universal causal mechanism. Post-L2 is a diagnostic baseline; the FastV comparisons show that RBM is not uniformly strongest even among the tested methods. The hybrid remains exploratory, and shared loading failures limit the scope of the HF OCRBench results. Broader selector comparisons and direct text-region interventions would be needed for general claims about competitiveness or semantic mechanisms.

Conclusion

Selecting native visual units from pre-merger features is a useful training-free compression design for the tested text-dense workloads. Separating the deployed RBM implementation from a final-input control shows that the useful signal persists at a later pre-merger tap, while score aggregation, feature depth, and stream handling also matter to operational performance. The GQA reversal and FastV comparisons define a clear boundary: selection representation should be evaluated against the intended workload, alongside the token budget and the model’s native interface.

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